

Deep Reinforcement-Learning-Based Adaptive Traffic Signal Control with Real-Time Queue Lengths

Qi-Wei Sun¹, Shi-Yuan Han^{1,*}, Jin Zhou¹, Yue-Hui Chen¹, Kang Yao²

¹Shandong Provincial Key Laboratory of Network Based Intelligent Computing,
University of Jinan, Jinan, 250022, China

² Shandong Provincial Network Security and Informationization Technology Center, Jinan, 250002, China
SunQwJimson@gmail.com, {ise_hansy, ise_zhouj, yhchen}@ujn.edu.cn, vipbaodao@163.com

*Corresponding Author. Email: ise_hansy@ujn.edu.cn

Abstract—The reinforcement learning (RL) with deep neural network, as a data-driven approach, is promising for adaptive traffic signal control (ATSC) in traffic scenarios. The majority of the existing studies focus on designing efficient agents and policy optimization for ATSC, but neglect to observe more detailed states of the environment. In this paper, an adaptive traffic signal control strategy, named as A2C_RTQL, is proposed for scheduling the traffic signal in an intersection, by combining the real-time lane-based queue lengths with deep RL agent. First, the Lighthill-Whitham-Richards (LWR) shockwave theory is employed for obtaining the real-time queue lengths in each lane. After that, by defining the obtained queue lengths as the inputs, A2C_RTQL strategy is designed for traffic signal control based on the advanced actor-critic (A2C) agent, where the lanes are divided into multiple parallel environments based on the phases of traffic signal. Simulation results demonstrate the optimality and efficiency of the proposed strategy compared with other methods in SUMO under simulated peak-hour traffic dynamics.

Index Terms—traffic signal control, real-time queue length, deep reinforcement learning, advanced actor-critic agent

I. INTRODUCTION

Due to the progressive increase of vehicles on the urban traffic network, the traffic congestion is becoming increasingly serious. Traffic signal control is by now the most direct and least costly way to maximize the traffic efficiency or ease the congestion. In general, most widely adaptive traffic signal control systems are designed based on the simplified traffic models [1]–[3], which relies on a large number of assumptions and empirical equations. However, traffic flows are usually complex and time varying in urban traffic scenarios.

The reinforcement learning (RL) has offered brilliant prospects in traffic signal control applications [4]–[6]. Different from the traditional model-driven approaches with heuristic assumptions and equations, RL methods obtain the optimal control strategy by directly fitting parametric models based on their empirical system of interaction with the complex traffic scenario. Nowadays, the deep neural networks (DNNs) [7] have advantageously been applied to increase the learning ability of RL on most complex tasks, thus being regarded as deep reinforcement learning (DRL) [8]. From value-based methods to policy based methods, researchers have studied various of DRL methods for ATSC problem. Paying attention to the value-based methods, [9]

introduces DRL to the field of traffic signal control by using neural networks to fit Q values; by employing DQN to learn the value function in [10], a learning-based traffic signal control algorithm is offered with neighborhood information sharing; by setting up a deep neural network to learn the Q -function of RL, a set of algorithms are advanced for signal timing plans via DRL in [11]. Focusing on policy-based methods, by employing the actor-critic method to learn the policy, an adaptive traffic signal controller is proposed in [12] in response to the changing traffic network state; in order to overcome the difficulties in dealing with a great amount of available traffic state information, a traffic light timing optimization strategy is proposed by using the deep deterministic policy gradient method to learn the policy in [13]. DRL provides an effective tool for solving ATSC problem under complex and time varying in urban traffic scenarios.

In addition to the above DRL methods by innovating learning algorithms, researchers have also been exploring different perception methods for the traffic state to improving DRL methods for ATSC. Traffic situation features (e.g. positions of vehicles) are converted into images in [14], and convolutional neural networks is employed to learn their representations. [15] proposes a DRQN model that integrates hidden states across multiple time steps into the DRL agent, but this obviously increases the dimensionality of the states to lead to the overfitting of the neural network. [16] employs a long short-term memory network (LSTM) in a DRL agent to extract dynamic traffic information. However, it is still very difficult to apply DRL methods in real-world scenarios because the selected features are limited by the real-world data collection methods and the communication delays. Fortunately, as an important indicator of traffic state, the queue lengths in real-time distance could be detected based on the geomagnetic sensors.

Linking the detected lane-based real-time queue lengths with DRL agent technologies, this paper proposes an adaptive traffic signal control strategy, named as A2C_RTQL. First, the real-time queue lengths in each lane are detected based on the Lighthill-Whitham-Richards (LWR) shockwave theory, which is integrated by using the Kalman filter. After that, by combining the advantages of value-based methods and policy-based methods, an advanced actor-critic (A2C)

agent is designed to scheduling the traffic signal, where the lanes are divided into multiple parallel environments based on the phases of traffic signal. Finally, the real high and low peaks of traffic flow are simulated to verify the effectiveness and feasibility of the proposed algorithm in the part of simulation experiment.

The remainder of the paper is organized as follows: The detection of real-time queue lengths are given in Section II. The design of A2C_RTQL is offered detailed in Section III. Section IV gives the simulation results. Some concluding remarks are presented in Section V.

II. REAL-TIME DETECTION OF QUEUE LENGTHS

The queue length in intersection is an important indicator to measure the efficiency of traffic control. In this section, the lane-based queue lengths are detected based on the collecting data from the geomagnetic sensors.

At present, the geomagnetic sensors are mainly divided into two types: (1) Passage detector: The geomagnetic vehicle detector, embedded in the road between two intersections, is used to detect the traffic flow and average speed in the lane; (2) Presence detector: The geomagnetic vehicle detector embedded, in front of the stop line in the intersection, is used to detect whether there are vehicles waiting to pass the intersection. Generally speaking, if the time step of calculating the queue distancelengths is too long, the signal light control will not be manipulated accordingly; if it is too short, the actions of RL will not be motivated on time due to the high computational cost and the long communication delay. In this paper, the time step t of detecting the queue lengths sets as 5 seconds.

The LWR shockwave theory [17] is used to detect the length of queue. The theory compares the traffic flow to a continuous fluid, in which the principles of fluid mechanics is employed to abstract a shock wave from the change of traffic density. Under the assumption that all of vehicles start queuing when their speed is 0, the relationship between the speed and density of traffic flow is obtained by analyzing the shock wave. The LWR shockwave theory is descried as follows more detailed.

First, the speeds of the queue-forming wave $w_{1,i}(t)$ and the residual queue-forming wave $w_{2,i}(t)$ during time step t are obtained as:

$$\begin{aligned} w_{1,i}(t) &= \frac{q_{p,i}(t)}{k_j - k_{p,i}(t)}, \\ w_{2,i}(t) &= \left| \frac{q_m - q_{p,i}(t)}{k_m - k_{p,i}(t)} \right| = \frac{q_m}{k_m}, \end{aligned} \quad (1)$$

where $q_{p,i}(t)$ is the vehicle arrival rate of the vehicles passing the passage detector in lane i and entering the intersection; $q_{p,i}(t)$ is first set to 0.9, and then updates by using the video detector of the intersection; k_j is the jam density; q_m and k_m are saturation volume and optimal density. $k_{p,i}(t)$ is the density of the traffic flow in lane i and can be obtained by dividing $q_{p,i}(t)$ by $\bar{v}_p(t)$. Then the queue length from $(n-1)$ -th to the n -th step is obtained as:

$$L_i(t-1, t) = \frac{5q_{p,i}(t)}{k_j - k_{p,i}(t)}. \quad (2)$$

When the signal turns to red at time step t , the residual queue length during the queue-discharging period is determined as:

$$L_{res,i}(t) = k_m \left(\frac{L_i(r)}{k_m} - \sum_{x=m}^t 5w_{2,i}(x) \right) + L_i(r, t-1), \quad (3)$$

where $L_{r,i}$ is the queue length of lane i at time step r (the end of the last red signal). Then the real-time queue length in lane i at time step t is detected as:

$$L_i(t) = L_i(t-1, t) + L_{res,i}(t). \quad (4)$$

III. ADAPTIVE TRAFFIC SIGNAL CONTROL STRATEGY: A2C_RTQL

A. Framework of A2C_RTQL Strategy

The basic framework of the proposed A2C_RTQL strategy for an intersection is introduced in this subsection, which combines the real-time calculation of lane-based queue lengths with the A2C agent for traffic signal control. The structure of the proposed A2C_RTQL strategy is shown in Fig. 1.

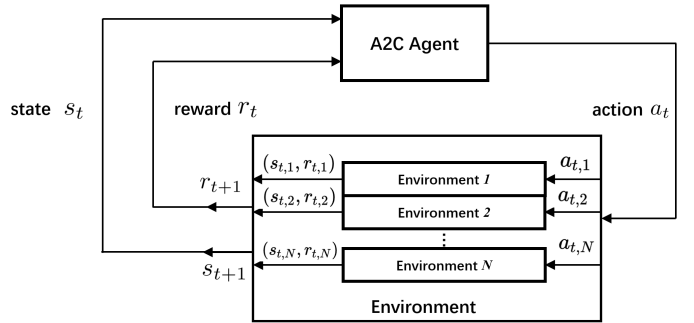


Fig. 1. The structure of the proposed A2C_RTQL strategy.

At any time step t , the flow of the proposed strategy is as follows: it is based on the detected real-time queue lengths that the agent first observes the current traffic state s_t and calculates the current reward r_t . After that, based on the obtained s_t and r_t , the action a_t is obtained and executed by the agent. Thus the feedback from the environment is received and the state s_{t+1} and reward r_{t+1} for the next step are obtained.

What's follows, the components of the proposed A2C_RTQL method will be described more detailed, including the environment, the state, the action, the reward function, and the A2C agent.

B. Environment

The proposed A2C_RTQL strategy divides all lanes in an intersection into multiple parallel environments based on signal lights, which is shown in Fig. 2.

Each environment interacts with the agent independently, that is, the agent observes each environment independently, and outputs a group of actions. Meanwhile, each environment corresponds to a group of actions, and the state $s_{t+1,j}$ and reward $r_{t+1,j}$ returned by the corresponding j -th environment are also grouped. So the the current traffic state $s_t = (s_{t,1}, s_{t,2}, s_{t,3}, \dots, s_{t,J})$, the action $a_t = (a_{t,1}, a_{t,2}, a_{t,3}, \dots, a_{t,J})$, and the reward $r_t =$

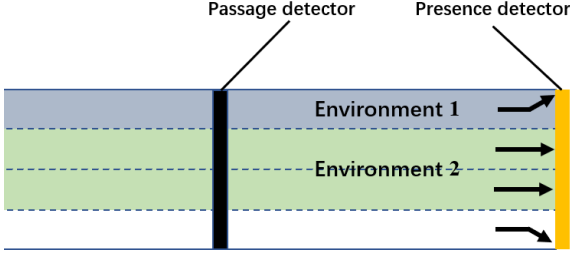


Fig. 2. The structure of the multiple parallel environments.

$(r_{t,1}, r_{t,2}, r_{t,3}, \dots, r_{t,J})$, where J is the number of environments.

C. State Definition

The state in the j -th environment at time step t is defined as

$$s_{t,j} = \sum_{i=1}^n L_i(t), \quad (5)$$

where $L_i(t)$ is the number of waiting vehicles in the i -th incoming lane of the j -th environment, and n is the number of lanes in the j -th environment.

D. Action Space

In this paper, the actions in A2C_RTQL consist in keeping the current signal phase and changing to the next signal phase.

In a standard intersection, the set of signal phases is $PhaseSpace = \{NSG, NSLG, EWG, EWLG\}$, including the north-south straight phase, the south-north left phase, the east-west straight phase, and the east-west left phase. The right-turning vehicles at the intersection are not constrained by the traffic signal. While the agent observes a traffic state, it will select an optimal action a according to the policy. When $a = 0$, the agent will keep the current phase; when $a = 1$, the agent will switch to the next phase. The minimum duration of each phase is set to 10s. For safety's sake, there is a 4s long yellow light between the green light and the red light. The set of signal phases and the definition of action are shown in Fig. 3.

E. Reward Function

As the key to the convergence of DRL model, the reward function $r_{t,j}$ is used to measure the quality of the action and provides the feedback for updating the action. In this thesis, the reward function $r_{t,j}$ is defined as the difference between the queue distances lengths of vehicles in all the lanes at the neighboring time steps. This definition of reward function is defined as:

$$r_{t,j} = L_j(t+1) - L_j(t), \quad (6)$$

where $L_j(t)$ is the sum of the queue length in the j -th environment at time t ; $L_j(t+1)$ is the sum of the queue length in the j -th environment at time step $(t+1)$.

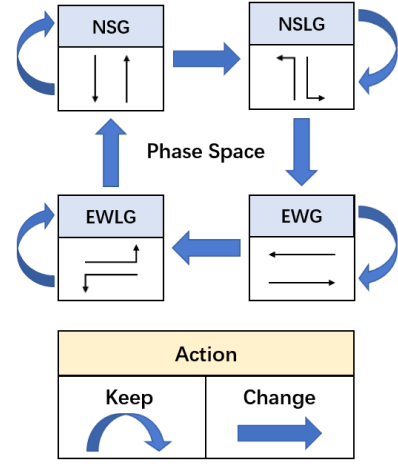


Fig. 3. The set of signal phases and the definition of action.

F. The A2C agent

By combining the advantages of value-based methods and policy-based methods, A2C agent [18] uses multiple workers to support the creating multiple parallel environments based on the phases of traffic signal.

For multiple parallel environments, as shown in Fig. 4, a north-south straight phase could be designated as one environment in an intersection. Corresponding, the other environments could be designated in the proposed A2C_RTQL strategy.

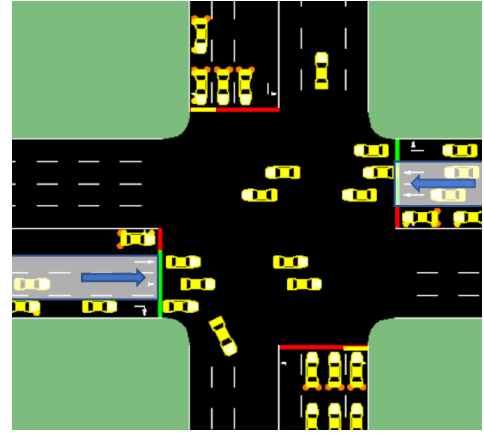


Fig. 4. One environment in multiple parallel environments.

Meanwhile, the structure of A2C agent is shown in Fig. 5.

The parallel training is used in the proposed A2C_RTQL strategy, where the entire agent consists of a master node and multiple parallel independent node called workers. Each worker and the master contains an Actor-Critic network and each worker interacts independently with its own environment to obtain independent states s and reward r . To lessen the load of heavy computation in the master node as much as possible, some computations are assigned to the worker nodes. Each worker keeps a copy of the current global critic. After collecting the data of a trajectory, the worker computes gradients $(\Delta_\psi, \Delta_\theta)$ for both the actor and critic. Instead of updating the local actor and critic, the gradients $(\Delta_\psi, \Delta_\theta)$ are sent back to the master. The task of the coordinator is then

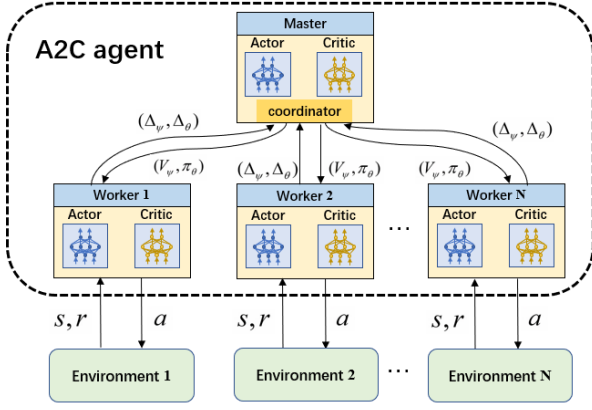


Fig. 5. The structure of A2C agent.

to collect and aggregate the gradients from all the workers and perform a simple update on the global models. Both the updated actor and critic (V_ψ, π_θ) are then synchronized again with all the workers. In A2C agent, the squared temporal difference (TD) error is used as the loss for the learning on the critic, and policy gradient with TD error is used to update the actor. Based on the above statements, the outlines of the master is given in Algorithm 1 and the workers in Algorithm 2.

Algorithm 1 Master

Require: step size η_ψ and η_θ , set of worker nodes W

Input: Initial policy parameters θ_0 , value function parameters ψ_0

- 1: Initialize $\theta = \theta_0$ and $\psi = \psi_0$
 - 2: **for** $t = 0, 1, 2, \dots$ **do**
 - 3: $(g_\psi, g_\theta) = 0$
 - 4: **for** $worker$ in W **do**
 - 5: $(g_\psi, g_\theta) = (g_\psi, g_\theta) + worker(V_\psi^{\pi_\theta}, \pi_\theta)$
 - 6: **end for**
 - 7: $\psi = \psi - \eta_\psi g_\psi; \theta = \theta - \eta_\theta g_\theta$
 - 8: **end for**
-

Algorithm 2 Worker

Require: Reward discount factor γ ,
the length of trajectory L

Input: Value function $V_\psi^{\pi_\theta}$, policy π_θ

- 1: Run policy π_θ for L time steps, collect S_t, A_t, R_t, S_{t+1}
 - 2: Estimate advantages $\hat{A} = R_t + \gamma V_\psi^{\pi_\theta}(S_{t+1}) - V_\psi^{\pi_\theta}(S_t)$
 - 3: $J(\theta) = \sum_t \log \pi_\theta(A_t | S_t) \hat{A}$
 - 4: $J_{V_\psi^{\pi_\theta}}(\psi) = \sum_t \hat{A}^2$
 - 5: $(g_\psi, g_\theta) = (\nabla J_{V_\psi^{\pi_\theta}}(\psi), \nabla J(\theta))$
 - 6: Return (g_ψ, g_θ)
-

IV. SIMULATION EXPERIMENTS

This section first introduces the simulation environment and the relevant parameter settings. After that, by using SUMO simulation software, the effectiveness of the proposed A2C_RTQL strategy is measured by comparing with the fixed timing control algorithm [19], the self-organizing traffic lights method [20], and A2C algorithm [18], respectively.

A. Simulation Environment and Parameter Settings

Based on the microscopic traffic simulation platform SUMO, the experiments are conducted on an Intel Core i9-9900K CPU taken as the hardware environment. The Traffic Control Interface module provided in SUMO is used to interact with the online simulation platform so that the traffic state in real time is obtained and the signal control strategy is adjusted properly. The scenarios are created by using the gym package. The model of the proposed A2C_RTQL strategy is implemented by using the deep learning framework Keras.

The simulation experiments are carried out for a single intersection. The intersection consists of four perpendicular roads with 500m and eight lanes in each direction. Each lane on the furthest left is the lane for turning left. The middle of two lanes are straight lanes and each on the furthest right is the lane for turning right.

a) *Traffic Flow Settings:* Assume that the generation of vehicles obeys the Weber distribution. The probability density function is defined as:

$$f(x; \lambda, m) = \begin{cases} \frac{m}{\lambda} \left(\frac{x}{\lambda}\right)^{m-1} e^{-(x/\lambda)^m} & x \geq 0 \\ 0 & x < 0 \end{cases}, \quad (7)$$

where $\lambda = 1$ is the scale parameter; $m = 2$ is the shape parameter. By simulating a real-life peak-to-peak situation, the maximum of vehicles enter the road network at a certain. Vehicles entering the traffic network from any entrance have 60% probability of going straight, 20% probability of turning left, and 20% probability of turning right. The vehicles are 5m long with a minimum spacing of 2.5m between vehicles. Meanwhile, the acceleration is 12m/s and the vehicle enters the network at a speed of 36km/h with a maximum speed of 50km/h. The distribution of the vehicle speed conforms to a Gaussian distribution and its probability density function is defined as:

$$f(x; \mu, \sigma) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(x-\mu)^2}{2\sigma^2}} & x \geq 0 \\ 0 & x < 0 \end{cases}, \quad (8)$$

where $\mu=40$ and $\sigma=5$.

b) *Hyperparameter Settings:* The numbers of training episodes and the environments set as 100 and 4, respectively. The model are updated every 5 steps. The Adam optimizer is used, where the learning rate is 0.001 and the batch size is 80. The squared temporal difference (TD) error is used as the loss function for the critic, and the policy gradient of the TD error is used to update the actor.

B. Simulation Results and Analyses

The simulation duration of the experiment is 5000s and the number of vehicles that enter the road network is 1500. Random seeds are used to generate the traffic flow data. The average waiting time, the average number of stops and the average cumulative rewards of the vehicles are chosen as the performance indicators. The following benchmark methods are employed to verify the validity of the proposed A2C_RTQL strategy.

(1) *FTC Algorithm:* FTC predefines a set of timing schemes based on the classical Webster's timing method and is widely used in real traffic scenarios.

(2) SOTL Algorithm: Based on the number of vehicles, this method works out how long the lights should stay green in order to relieve any congestion.

(3) A2C Algorithm: A2C algorithm uses the same agent as the proposed method A2C_RTQL. The difference is that A2C only use the cumulative delay of the first vehicle as the state. The remainder including the hyperparameter settings are the same as the proposed A2C_RTQL strategy.

The comparison results of the cumulative rewards between A2C_RTQL strategy and A2C algorithm are shown in Fig. 6 in the training process under the same traffic flow conditions. It is obviously that the variation tendency and values are similar under two strategies. It means that the addition of the calculation of queue lengths will not reduce the convergence speed of the methods and influence the stability of the methods. Meanwhile, the comparison results of the average waiting time are shown in Fig. 7 under A2C_RTQL, A2C, FTC, and SOTL strategies, respectively. It can be seen that, in the initial stage of training, the agent has not yet learned the reasonable control strategy for A2C_RTQL so that the average waiting time increases significantly. As the training proceeds, the performance of A2C_RTQL and A2C gradually exceeds those of FTC and SOTL.

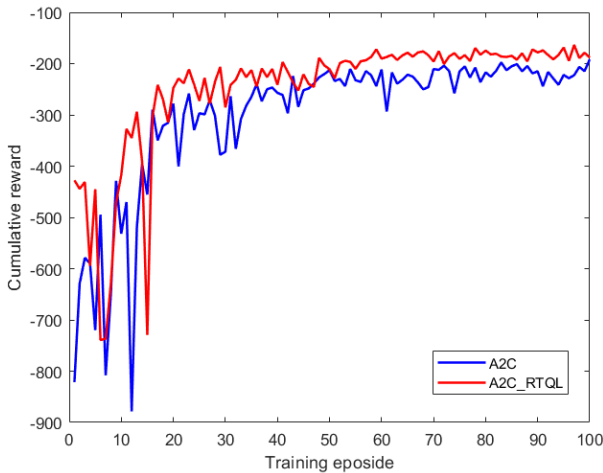


Fig. 6. Comparison curves of cumulative reward under A2C_RTQL and A2C strategies.

For showing the numeric comparison results among the proposed A2C_RTQL strategy and the employed benchmark methods, 20 sets of randomly generated traffic data are generated and tested. Considering the main indicators of the cumulative reward and the average waiting time, the corresponding results of average values are shown in Table I. At the same time, selecting the travel time, the fuel consumption, and the CO_2 emissions as the secondary indicators, the corresponding results of average values are shown in Table II. Observing the main indicators, it is found that the performance of A2C_RTQL is better than those of FTC, SOTL and A2C. Specially compared with A2C, the average waiting time of A2C_RTQL is reduced by 6.52%, and the cumulative reward is increased by 5.16%. Meanwhile, for the secondary indicators, the improvement under the proposed A2C_RTQL strategy is also marked.

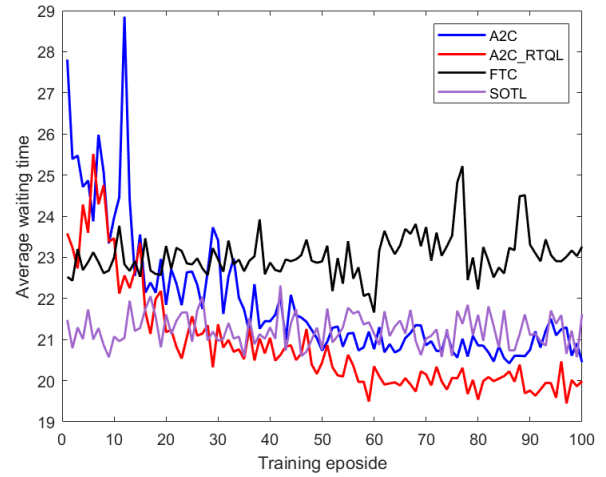


Fig. 7. Comparisons curves of average waiting time under different control strategies.

TABLE I
COMPARISONS OF THE MAIN INDICATORS

Algorithms	The Main Indicators	
	Waiting Time (s)	Cumulative Reward
FTC	27.50	\
SOTL	24.16	
A2C	22.99	-202.14
A2C_RTQL	21.49	-194.71

Based on the above simulation results and analyses, the proposed A2C_RTQL strategy could reduce the vehicle waiting time, decrease the air pollution and alleviate the traffic congestion effectively.

V. CONCLUSION

This paper proposes a strategy for the adaptive traffic signal control by integrating DLR with detection method of lane-based queue lengths. The lengths of the real-time lane-based queue were obtained based on the LWR shockwave theory. Based on the designed A2C agent, the A2C_RTQL strategy was proposed to adjust the traffic signal adaptively, in which the lanes were divided into multiple parallel environments according to the released signal phases in an intersection. The effectiveness of the proposed A2C_RTQL strategy was proved by comparing with three benchmark methods. For the practical deployment of the proposed algorithm, our further work will focus on the improvement of ensuring the robustness on the noise and reducing the computational cost and the communication delay.

TABLE II
COMPARISONS OF THE SECONDARY INDICATORS

Algorithms	The Secondary Indicators		
	Travel Time (s)	Fuel consumption (ml)	CO_2 emissions (g)
FTC	116.57	89.33	213.68
SOTL	109.20	77.56	185.52
A2C	104.88	76.29	182.49
A2C_RTQL	102.67	75.30	180.12

ACKNOWLEDGMENT

This research was funded by the National Natural Science Foundation of China under Grants 61903156 and 61873324, the Natural Science Foundation of Shandong Province for Key Project under Grant ZR2020KF006, the Natural Science Foundation of Shandong Province under Grant ZR2019MF040, the University Innovation Team Project of Jinan under Grant 2019GXRC015, and the State Scholarship Fund of the China Scholarship Council.

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